

Bayesian decision theory

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Discrimination function

Discriminant function $g_i(\mathbf{x})$ is used to partition $\mathbb{R}^d$ space into c classes, $i = 1, 2, 3, \dots, c$

Decision Rule

Assign $\mathbf{x}$ to class ω_j if

$$g_j(\mathbf{x}) > g_i(\mathbf{x}) \quad \forall i = 1, 2, \dots, c \text{ \& } i \neq j$$

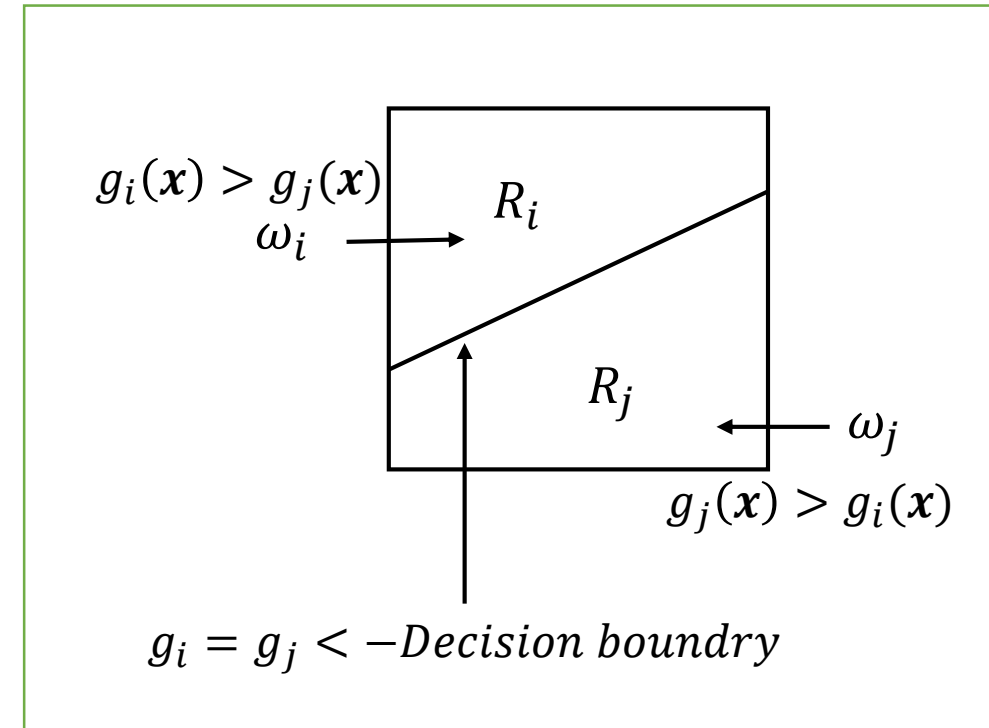
Decision Boundary

The decision boundary between classes ω_i and ω_j is defined by ω

$$g_j(\mathbf{x}) = g_i(\mathbf{x}) \leftarrow \text{Decision Boundary}$$
$$g(\mathbf{x}) = g_j(\mathbf{x}) - g_i(\mathbf{x}) = 0$$

Linear Discriminant Function

$$g_i(\mathbf{x}) = \mathbf{w}_i^T \mathbf{x} + w_{i0}$$



Bayesian decision theory

Assume, two class problem

e.g., In a conveyer belt, there are 2 types of fishes Salmon and Sea bass.

Let, Salmon Fish belongs to class ω_1 , Sea bass Fish belongs to class ω_2 ,

Probability of Salmon = $p(\omega_1)$ known
Probability of Sea bass = $p(\omega_2)$ known

} Prior Probability

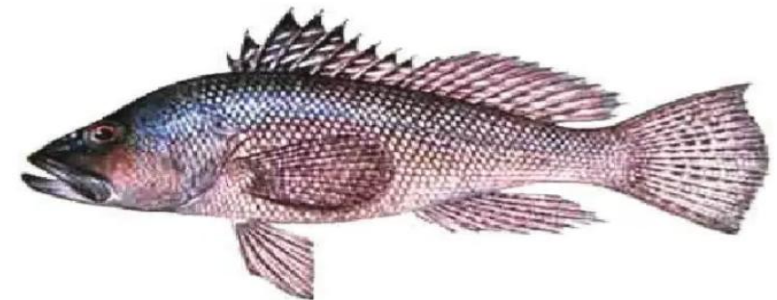
We can make simple decision rule:-

If $p(\omega_1) > p(\omega_2)$, then decide class ω_1

If $p(\omega_2) > p(\omega_1)$, then decide class ω_2



ω_1 Salmon



ω_2 Sea bass

Bayesian decision theory

Key idea is to look for discriminant features ($\mathbf{x}$)

I can find out the probabilistic measure or probability density function (PDF) of variable $\mathbf{x}$ for object which belongs to class ω_1 and ω_2 separately.

$$p(\mathbf{x}|\omega_1) \text{ and } p(\mathbf{x}|\omega_2)$$



Class conditional PDF

$p(\mathbf{x}|\omega_i)$: how the class generates data

$p(\omega_i)$: how common the class is

$p(\omega_i|\mathbf{x})$: how confident we are after seeing data

Our objective is to calculate:-

$$p(\omega_1|\mathbf{x}) \text{ and } p(\omega_2|\mathbf{x})$$



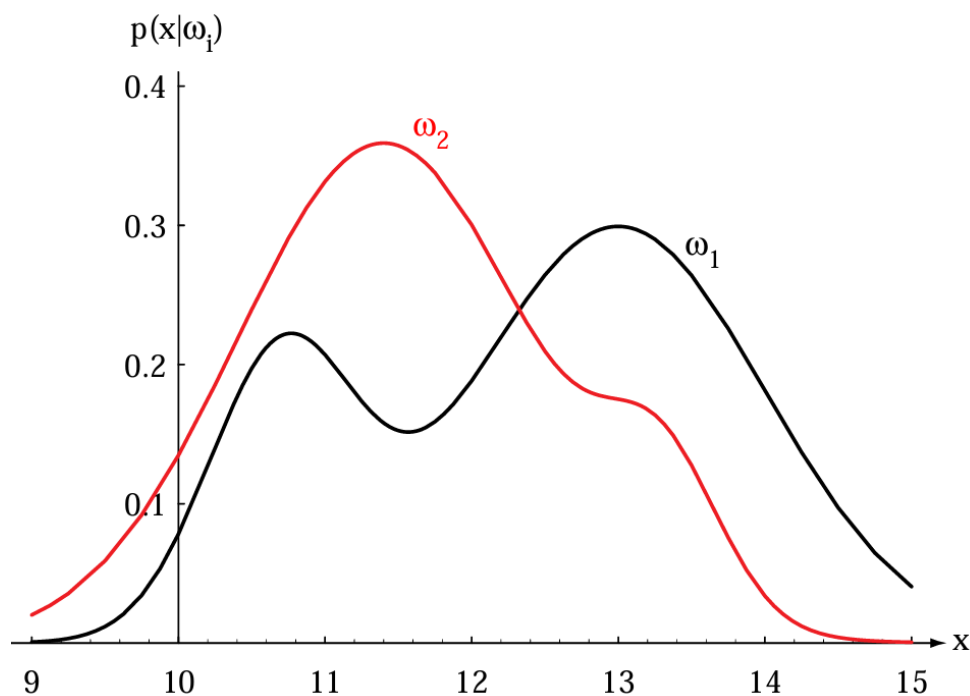
Posterior probability

Joint probability density function,

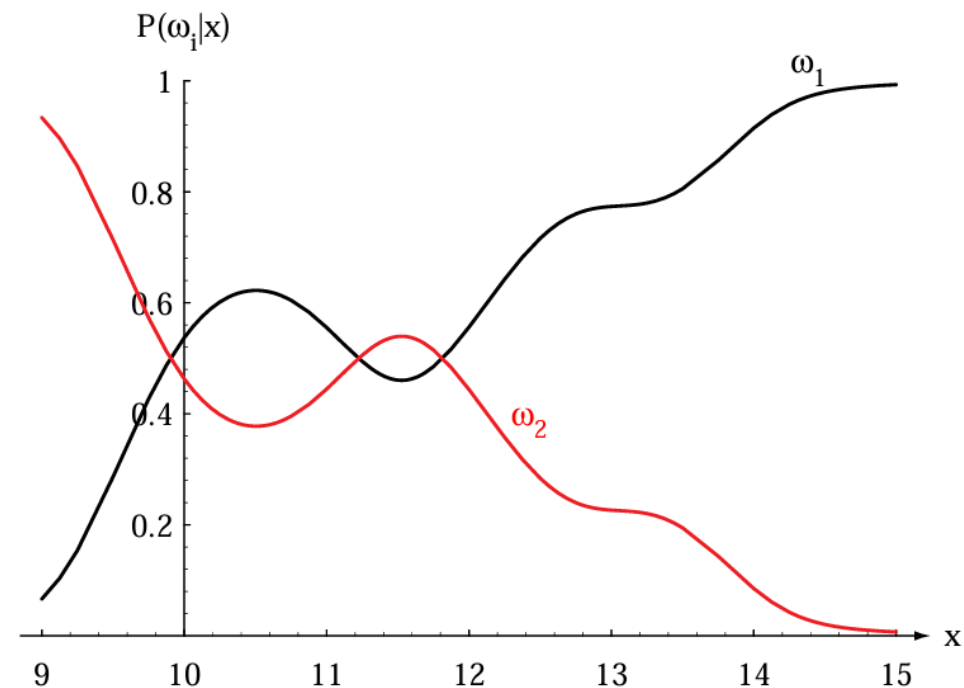
$$p(\omega_i, \mathbf{x}) = p(\omega_i|\mathbf{x}) p(\mathbf{x})$$

$$= p(\mathbf{x}|\omega_i) p(\omega_i)$$

Bayesian decision theory



Hypothetical class-conditional probability density functions show the probability density of measuring a particular feature value x given the pattern is in category ω_i . If x represents the length of a fish, the two curves might describe the difference in length of populations of two types of fish. Density functions are normalized, and thus the area under each curve is 1.0.



Posterior probabilities for the particular priors $P(\omega_1) = 2/3$ and $P(\omega_2) = 1/3$ for the class-conditional probability densities shown in previous figure. Thus in this case, given that a pattern is measured to have feature value $x = 14$, the probability it is in category ω_2 is roughly 0.08, and that it is in ω_1 is 0.92. At every x , the posteriors sum to 1.0.

Bayesian decision theory

$$\Rightarrow p(\omega_i|\mathbf{x}) p(\mathbf{x}) = p(\mathbf{x}|\omega_i) p(\omega_i)$$

$$\Rightarrow p(\omega_i|\mathbf{x}) = \frac{p(\mathbf{x}|\omega_i) p(\omega_i)}{p(\mathbf{x})}$$

$$\text{Posterior} = \frac{\text{Likelihood} \times \text{Prior}}{\text{Evidence}}$$

$$p(\omega_i|\mathbf{x}) = \frac{p(\mathbf{x}|\omega_i) p(\omega_i)}{p(\mathbf{x})}$$

Bayes rule

$$p(\mathbf{x}) = \sum_{i=1}^2 p(\mathbf{x}|\omega_i) p(\omega_i)$$

If $p(\omega_1|\mathbf{x}) > p(\omega_2|\mathbf{x})$, then decide class ω_1

If $p(\omega_2|\mathbf{x}) > p(\omega_1|\mathbf{x})$, then decide class ω_2

Bayesian decision theory

Decision Rule (MAP)

If $p(\mathbf{x}|\omega_1)p(\omega_1) > p(\mathbf{x}|\omega_2)p(\omega_2)$, then decide ω_1

Otherwise, decide ω_2

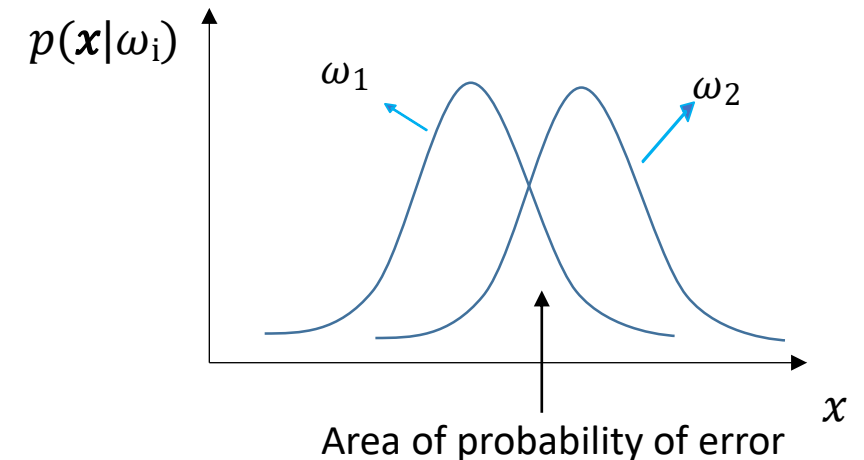
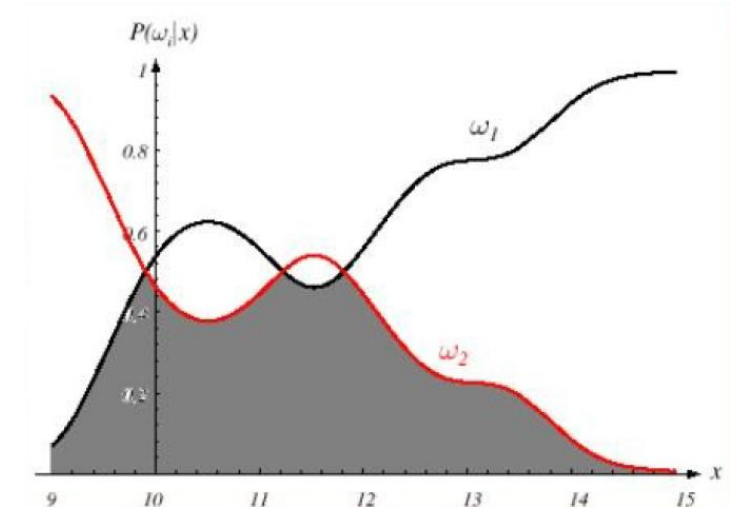
If equality holds, the classifier is at the decision boundary.

Conditional Probability of Error

$$p(\text{error}|\mathbf{x}) = \begin{cases} p(\omega_1|\mathbf{x}), & \text{if decide } \omega_2 \\ p(\omega_2|\mathbf{x}), & \text{if decide } \omega_1 \end{cases}$$

Minimum Error Rule (0–1 loss)

decide ω_1 if $p(\omega_1|\mathbf{x}) > p(\omega_2|\mathbf{x})$, otherwise ω_2



Bayesian decision theory

Avg. Probability of error

$$p(\text{error}) = \int_{-\infty}^{\infty} p(\text{error}|\mathbf{x}) d\mathbf{x} = \int_{-\infty}^{\infty} p(\text{error}|\mathbf{x}) p(\mathbf{x}) d\mathbf{x}$$

Conditional Probability of Error (0-1 loss)

$$p(\text{error}|\mathbf{x}) = \min [p(\omega_1|\mathbf{x}), p(\omega_2|\mathbf{x})] \quad (\text{for 0-1 loss})$$

Posterior Probability

$$p(\omega_i|\mathbf{x}) = \frac{p(\mathbf{x}|\omega_i) p(\omega_i)}{\sum_{i=1}^2 p(\mathbf{x}|\omega_i) p(\omega_i)} = \frac{p(\mathbf{x}|\omega_i)p(\omega_i)}{p(\mathbf{x})}$$

Decision Rule (minimum error)

If $p(\omega_1|\mathbf{x}) > p(\omega_2|\mathbf{x})$, then decide class ω_1
otherwise, decide class ω_2

Generalized Bayes classifier

- Use more than two states of nature
- More than one feature
- More action to consider
- Loss function

Classes

$c \rightarrow$ No. of classes; $\{\omega_1, \omega_2, \dots, \omega_c\}$

Actions

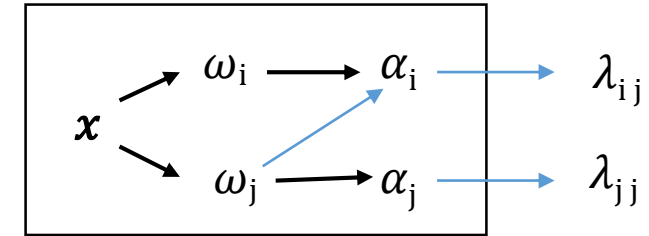
$\{\alpha_1, \alpha_2, \dots, \alpha_a\}$

Feature vector

$\mathbf{x} = [x_1, x_2, \dots, x_d]^T$

Loss function

$\lambda(\alpha_i | \omega_j)$, loss is occurred for taking action α_i when true state of class is ω_j .



Risk function

$$R_i = R(\alpha_i|\mathbf{x}) = \sum_{j=1}^c \lambda(\alpha_i|\omega_j) p(\omega_j|\mathbf{x})$$

↑
Risk function/conditional risk/expected loss.

For simplicity, $\lambda(\alpha_i|\omega_j) = \lambda_{ij}$

In general,

$$R(\alpha_i|\mathbf{x}) = \sum_{j=1}^c \lambda_{ij} p(\omega_j|\mathbf{x})$$

Total risk for all $\mathbf{x}$

$$R = \int_{-\infty}^{\infty} R_i p(\mathbf{x}) d\mathbf{x}$$

If $p(\omega_i|\mathbf{x}) > p(\omega_j|\mathbf{x}) \forall i \neq j$
decide action α_i

Minimum Risk classifier

$$R(\alpha_i|\mathbf{x}) < R(\alpha_j|\mathbf{x}) \forall i \neq j$$

For two class problem

For action α_1 , $R(\alpha_1|\mathbf{x}) = \lambda_{11} p(\omega_1|\mathbf{x}) + \lambda_{12}p(\omega_2|\mathbf{x})$

$$\begin{array}{ccc} \downarrow & & \downarrow \\ \lambda(\alpha_1|\omega_1) & & \lambda(\alpha_1|\omega_2) \end{array}$$

For action α_2 , $R(\alpha_2|\mathbf{x}) = \lambda_{21} p(\omega_1|\mathbf{x}) + \lambda_{22}p(\omega_2|\mathbf{x})$

$$\begin{array}{ccc} \downarrow & & \downarrow \\ \lambda(\alpha_2|\omega_1) & & \lambda(\alpha_2|\omega_2) \end{array}$$

If $R(\alpha_1|\mathbf{x}) < R(\alpha_2|\mathbf{x})$, then in favour of α_1

If $R(\alpha_1|\mathbf{x}) > R(\alpha_2|\mathbf{x})$, then in favour of α_2

$$\lambda_{21} p(\omega_1|\mathbf{x}) + \lambda_{22}p(\omega_2|\mathbf{x}) > \lambda_{11}p(\omega_1|\mathbf{x}) + \lambda_{12}p(\omega_2|\mathbf{x})$$

for decision in favor of w_1 or action α_1

$$(\lambda_{21} - \lambda_{11})p(\omega_1|\mathbf{x}) > (\lambda_{12} - \lambda_{22}) p(\omega_2|\mathbf{x})$$

$$p(\omega_1|\mathbf{x}) / p(\omega_2|\mathbf{x}) > (\lambda_{12} - \lambda_{22}) / (\lambda_{21} - \lambda_{11})$$

$$p(\mathbf{x}|\omega_1) / p(\mathbf{x}|\omega_2) > (\lambda_{12} - \lambda_{22}) / (\lambda_{21} - \lambda_{11}) p(\omega_2) / p(\omega_1) \leftarrow T \text{ (Threshold, independent of } \mathbf{x})$$

If both, $(\lambda_{21} - \lambda_{11}) > 0$ and $(\lambda_{12} - \lambda_{22}) > 0$

Special case (0–1 loss)

$$\lambda_{11}=\lambda_{22}=0, \quad \lambda_{12}=\lambda_{21}=1$$

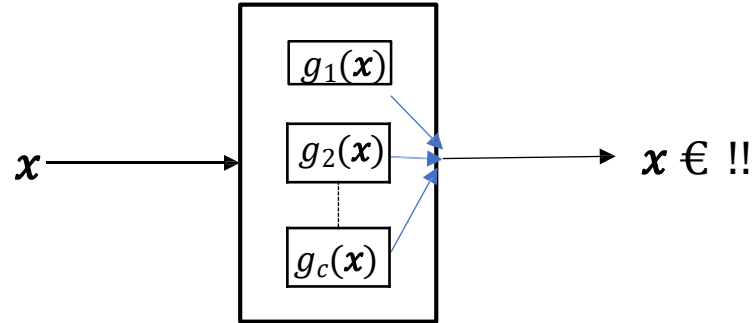
$$(\lambda_{12} - \lambda_{22})/(\lambda_{21} - \lambda_{11}) = 1$$

and the rule **reduces to**:

$$p(\omega_1|\mathbf{x}) > p(\omega_2|\mathbf{x})$$

This is the **minimum error rate / Maximum A Posteriori (MAP) classifier**

Multi-category class



$c = \text{No. of classes}$ $g(x) = \text{discriminant function}$

$\{\omega_1, \omega_2, \dots, \omega_c\}$, are c no of classes

$g_i(x); i = 1, 2, \dots, c.$

If $g_i(x) > g_j(x) \quad \forall j \neq i$ decide $x \in \omega_i$.

Minimum risk classifier

Maximum discriminant function corresponds to minimum risk

$$g_i(\mathbf{x}) = -R(\alpha_i|\mathbf{x})$$

For 0–1 loss,

$$R(\alpha_i|\mathbf{x}) = 1 - p(\omega_i|\mathbf{x}) \quad \text{Dropping constants (monotonic transformation)}$$

$$\text{Then, } g_i(\mathbf{x}) = -(1 - p(\omega_i|\mathbf{x})) = p(\omega_i|\mathbf{x})$$

If $f(\cdot)$ is monotonic increasing, then $\arg\max_i g_i(\mathbf{x}) = \arg\max_i f(g_i(\mathbf{x}))$

Minimum error rate classification:-

$$\longrightarrow p(\mathbf{x})$$

$$g_i(\mathbf{x}) = p(\omega_i|\mathbf{x}) = \frac{p(\mathbf{x}|\omega_i) p(\omega_i)}{\sum_{j=1}^2 p(\mathbf{x}|\omega_j) p(\omega_j)}$$

$$\propto p(\mathbf{x}|\omega_i) p(\omega_i)$$

Taking log (monotonic transformation),

$$g_i(\mathbf{x}) = \ln p(\omega_i|\mathbf{x}) = \ln p(\mathbf{x}|\omega_i) + \ln p(\omega_i)$$

Decision boundary between class i and j

Class ω_i } We have now two discriminant function $g_i(\mathbf{x})$ and $g_j(\mathbf{x})$
Class ω_j }

If $g_i(\mathbf{x}) > g_j(\mathbf{x})$ decide class ω_i }
If $g_i(\mathbf{x}) < g_j(\mathbf{x})$ decide class ω_j } Decision boundary $g_i(\mathbf{x}) - g_j(\mathbf{x}) = 0$

Single discriminant function:

$$\begin{aligned} g(\mathbf{x}) &= g_i(\mathbf{x}) - g_j(\mathbf{x}) = \ln p(\omega_i|\mathbf{x}) - \ln p(\omega_j|\mathbf{x}) \\ &= \ln p(\mathbf{x}|\omega_i) + \ln p(\omega_i) - \ln p(\mathbf{x}|\omega_j) - \ln p(\omega_j) \\ &= \ln \frac{p(\mathbf{x}|\omega_i)}{p(\mathbf{x}|\omega_j)} + \ln \frac{p(\omega_i)}{p(\omega_j)} \end{aligned}$$

Conditional Independence (Naïve Bayes)

Class conditional probabilities

$$p(\mathbf{x} | \omega_1) = \prod_{i=1}^d [p_i^{x_i} (1 - p_i)^{1-x_i}]$$
$$p(\mathbf{x} | \omega_2) = \prod_{i=1}^d [q_i^{x_i} (1 - q_i)^{1-x_i}]$$

- binary features $x_i \in \{0, 1\}$
- class-conditional Bernoulli model

$$p_i = p[x_i=1 | \omega_1]$$

$$q_i = p[x_i=1 | \omega_2]$$

Likelihood ratio

$$\frac{p(\mathbf{x} | \omega_1)}{p(\mathbf{x} | \omega_2)} = \prod_{i=1}^d \left[\left(\frac{p_i}{q_i} \right)^{x_i} \left(\frac{1 - p_i}{1 - q_i} \right)^{1-x_i} \right]$$

Taking ln both the side

$$\ln \frac{p(\mathbf{x} | \omega_1)}{p(\mathbf{x} | \omega_2)} = \ln \prod_{i=1}^d \left[\left(\frac{p_i}{q_i} \right)^{x_i} \left(\frac{1 - p_i}{1 - q_i} \right)^{1-x_i} \right]$$

Two category case

Discrimination function

$$g(\mathbf{x}) = \sum_{i=1}^d \left[x_i \ln \left(\frac{p_i}{q_i} \right) + (1 - x_i) \ln \left(\frac{1 - p_i}{1 - q_i} \right) \right] + \ln \frac{p(\omega_1)}{p(\omega_2)}$$

$$g(\mathbf{x}) = \sum_{i=1}^d \left[x_i \left(\ln \left(\frac{p_i}{q_i} \right) - \ln \left(\frac{1 - p_i}{1 - q_i} \right) \right) + \ln \left(\frac{1 - p_i}{1 - q_i} \right) \right] + \ln \frac{p(\omega_1)}{p(\omega_2)}$$

$$g(\mathbf{x}) = \sum_{i=1}^d \left[x_i \left(\ln \frac{p_i(1 - q_i)}{q_i(1 - p_i)} \right) \right] + \sum_{i=1}^d \ln \left(\frac{1 - p_i}{1 - q_i} \right) + \ln \frac{p(\omega_1)}{p(\omega_2)}$$

$$g(\mathbf{x}) = \sum_{i=1}^d w_i x_i + w_0$$

$$\text{where, } w_i = \ln \frac{p_i(1 - q_i)}{q_i(1 - p_i)}, w_0 = \sum_{i=1}^d \ln \left(\frac{1 - p_i}{1 - q_i} \right) + \ln \frac{p(\omega_1)}{p(\omega_2)}$$

$i = 1, 2, \dots, d$

- Decide ω_1 if $g(\mathbf{x}) > 0$ and ω_2 if $g(\mathbf{x}) < 0$
- If $p_i = q_i$, x_i gives no information about the state of nature and $w_i = 0$ (feature independence)
- If $p_i > q_i$, then $1 - p_i < 1 - q_i$, then the presence of feature x_i favors class ω_1 (ω_1 is positive) then decision will be in favour of ω_1 otherwise ω_2
- If $p(\omega_1) > p(\omega_2) \rightarrow$ increase bias w_{i0} then shifting the decision boundary in favour of ω_1 otherwise ω_2

Gaussian Naïve Bayes

Univariate Density:

$$p(x) = \frac{1}{\sqrt{2\pi}\sigma} \exp \left[-\frac{1}{2} \left(\frac{x-\mu}{\sigma} \right)^2 \right]$$

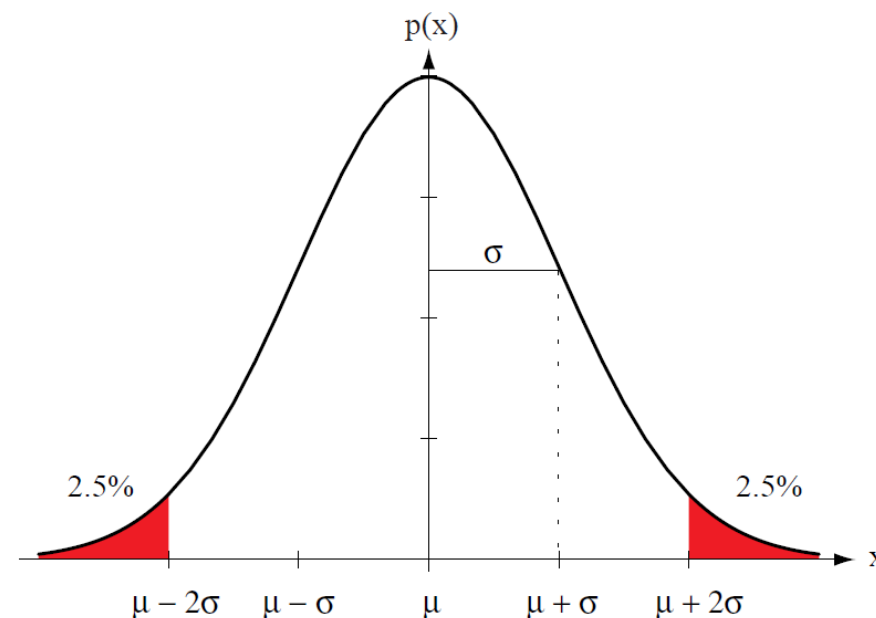
μ = expected value of x

$$\mu = E[x] = \int_{-\infty}^{\infty} xp(x)dx$$

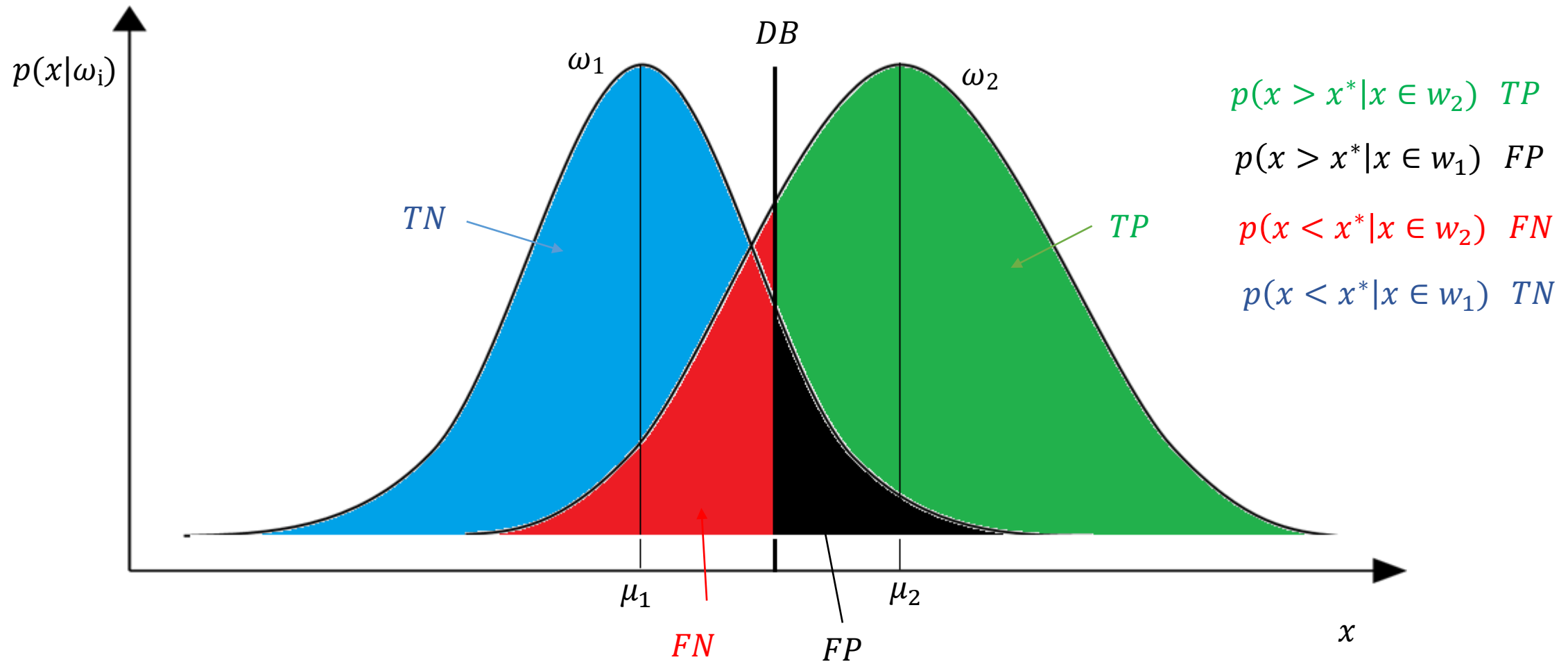
σ^2 = variance

$$\sigma^2 = E[(x - \mu)^2] = \int_{-\infty}^{\infty} (x - \mu)^2 p(x)dx \rightarrow N(\mu, \sigma^2)$$

$$p(x) \sim \mathcal{N}(\mu, \sigma^2)$$



Evaluation of a classifier based on likelihood



Multivariate Density

Multivariate normal density in d dimensions

$$P(\mathbf{x}) = \frac{1}{(2\pi)^{d/2} |\Sigma|^{1/2}} \exp \left[-\frac{1}{2} (\mathbf{x} - \boldsymbol{\mu})^t \Sigma^{-1} (\mathbf{x} - \boldsymbol{\mu}) \right]$$

$$\boldsymbol{\mu} = E(\mathbf{x}) = \int_{-\infty}^{\infty} \mathbf{x} p(\mathbf{x}) d\mathbf{x}$$

Σ = covariance matrix

$$\Sigma = E[(\mathbf{x} - \boldsymbol{\mu})(\mathbf{x} - \boldsymbol{\mu})^t] = \int_{-\infty}^{\infty} (\mathbf{x} - \boldsymbol{\mu})(\mathbf{x} - \boldsymbol{\mu})^t p(\mathbf{x}) d\mathbf{x}$$

i th component $\mu_i = E[x_i]$

$(i, j)^{th}$ component $\sigma_{i,j} = E[(x_i - \mu_i)(x_j - \mu_j)^t]$

Diagonal component, $\sigma_{i,i} = E[(x_i - \mu_i)^2] = \sigma_i^2$

$$\mathbf{p}(\mathbf{x}) \sim \mathcal{N}(\boldsymbol{\mu}, \Sigma)$$

$$\mathbf{x}^t = (x_1 \quad x_2 \quad \dots \quad x_d) \in \mathbb{R}^{1 \times d}$$

$$\boldsymbol{\mu}^t = (\mu_1 \quad \mu_2 \quad \dots \quad \mu_d) \in \mathbb{R}^{1 \times d}$$

$$\Sigma = \begin{pmatrix} \sigma_{11} & \sigma_{12} & \dots & \sigma_{1d} \\ \sigma_{21} & \sigma_{22} & \dots & \sigma_{2d} \\ \vdots & \vdots & \ddots & \vdots \\ \sigma_{d1} & \sigma_{d2} & \dots & \sigma_{dd} \end{pmatrix} \in \mathbb{R}^{d \times d}$$

$$(\mathbf{x} - \boldsymbol{\mu}) \rightarrow (d \times 1)$$

$$(\mathbf{x} - \boldsymbol{\mu})^t \rightarrow (1 \times d)$$

$$\Sigma \rightarrow (d \times d)$$

Discriminant Functions for the Normal Density

Multivariate Density

$$p(\mathbf{x} | \omega_i) = \frac{1}{(2\pi)^{d/2} |\Sigma_i|^{1/2}} \exp \left[-\frac{1}{2} (\mathbf{x} - \boldsymbol{\mu}_i)^t \Sigma_i^{-1} (\mathbf{x} - \boldsymbol{\mu}_i) \right]$$
$$\ln p(\mathbf{x} | \omega_i) = -\frac{1}{2} [(\mathbf{x} - \boldsymbol{\mu}_i)^t \Sigma_i^{-1} (\mathbf{x} - \boldsymbol{\mu}_i)] - \frac{d}{2} \ln 2\pi - \frac{1}{2} \ln |\Sigma_i|$$

We know, $g_i(\mathbf{x}) = \ln p(\omega_i | \mathbf{x}) = \ln p(\mathbf{x} | \omega_i) + \ln p(\omega_i)$

Discriminant Functions

$$g_i(\mathbf{x}) = -\frac{1}{2} [(\mathbf{x} - \boldsymbol{\mu}_i)^t \Sigma_i^{-1} (\mathbf{x} - \boldsymbol{\mu}_i)] - \frac{d}{2} \ln 2\pi - \frac{1}{2} \ln |\Sigma_i| + \ln p(\omega_i)$$

$$g_i(\mathbf{x}) = -\frac{1}{2} [(\mathbf{x} - \boldsymbol{\mu}_i)^t \Sigma_i^{-1} (\mathbf{x} - \boldsymbol{\mu}_i)] + \ln p(\omega_i) + c_i$$

$$\text{Where } c_i = -\frac{d}{2} \ln 2\pi - \frac{1}{2} \ln |\Sigma_i|$$

$$g_i(\mathbf{x}) = \underbrace{-\frac{1}{2} \mathbf{x}^t \Sigma_i^{-1} \mathbf{x}}_{\text{Non linear quadratic form}} + \frac{1}{2} \mathbf{x}^t \Sigma_i^{-1} \boldsymbol{\mu}_i + \frac{1}{2} \boldsymbol{\mu}_i^t \Sigma_i^{-1} \mathbf{x} - \frac{1}{2} \boldsymbol{\mu}_i^t \Sigma_i^{-1} \boldsymbol{\mu}_i + \ln p(\omega_i) + c_i$$

Non linear quadratic form

Let us examine the discriminant function and resulting classification for no. of special cases.

If $\Sigma_i = \Sigma$ then covariance matrix is same for all the classes but if is arbitrary. In that case, it will have no influence on decision boundary

Similarly, c_i will have no influence on decision boundary

$$g_i(\mathbf{x}) = (\boldsymbol{\mu}_i \Sigma^{-1})^t \mathbf{x} - \frac{1}{2} \boldsymbol{\mu}_i^t \Sigma^{-1} \boldsymbol{\mu}_i + \ln p(\omega_i)$$

$$g_i(\mathbf{x}) = w_i \mathbf{x} + w_{i0}$$

$$w_i = (\boldsymbol{\mu}_i \Sigma^{-1})^t \text{ and}$$

$$w_{i0} = \ln p(\omega_i) - \frac{1}{2} \boldsymbol{\mu}_i^t \Sigma^{-1} \boldsymbol{\mu}_i$$

Bivariate normal density function

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$p(\mathbf{x}) = \frac{1}{(2\pi)|\Sigma|^{1/2}} \exp \left[-\frac{1}{2} \left\{ \left(\frac{x_1 - \mu_1}{\sigma_1} \right)^2 + \left(\frac{x_2 - \mu_2}{\sigma_2} \right)^2 \right\} \right]$$

$$\Sigma = \begin{bmatrix} \sigma_1^2 & 0 \\ 0 & \sigma_2^2 \end{bmatrix} \quad \boldsymbol{\mu} = \begin{bmatrix} \mu_1 \\ \mu_2 \end{bmatrix}$$

$$p(\mathbf{x} | \omega_i) = \frac{1}{(2\pi)|\Sigma_i|^{1/2}} \exp \left[-\frac{1}{2} \left(\frac{x_1 - \mu_1}{\sigma_1} \right)^2 \right] \exp \left[\left(\frac{x_2 - \mu_2}{\sigma_2} \right)^2 \right]$$

It is squared Euclidean distance

Case 1: $\Sigma_i = \sigma^2 \mathbf{I}$

$$\Sigma = \begin{pmatrix} \sigma^2 & 0 & \cdots & 0 \\ 0 & \sigma^2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \sigma^2 \end{pmatrix} \in \mathbb{R}^{d \times d}$$

$\sigma_{11} = \sigma_{22} = \dots = \sigma_{dd}$

$$\Sigma_i = \sigma^2 \mathbf{I} \quad \mathbf{I} = \text{Identity matrix } (d \times d)$$

$\sigma_{i,j} = 0$, Off-diagonal elements of Σ are zero, implying statistical independence among features.

$$|\Sigma_i| = \sigma^2 \quad \left| \Sigma_i^{-1} \right| = \frac{1}{\sigma^2} \mathbf{I}$$

$$g_i(x) = -\frac{1}{2} [(x - \mu_i)^t \Sigma_i^{-1} (x - \mu_i)] - \underbrace{\frac{d}{2} \ln 2\pi - \frac{1}{2} \ln |\Sigma_i|}_{\text{Constant}} + \ln p(\omega_i)$$

Constant $\rightarrow$ Independent of i , so they are ignored.

Thus we obtain the simple discriminant functions

$$g_i(x) = -\frac{\|x - \mu_i\|^2}{2\sigma^2} + \ln p(\omega_i)$$

where $\|\cdot\|$ is the Euclidean norm that is,

$$\|x - \mu_i\|^2 = (x - \mu_i)^t (x - \mu_i)$$

Case 1: $\Sigma_i = \sigma^2 \mathbf{I}$

Expansion of the quadratic form $(x - \mu_i)^t(x - \mu_i)$ yields

$$g_i(x) = -\frac{1}{2\sigma^2} [x^t x - 2\mu_i^t x + \mu_i^t \mu_i] + \ln p(\omega_i)$$

which appears to be quadratic function of x . However, the quadratic term $x^t x$ is independent for all i , making it an ignorable additive constant. then,

$$g_i(x) = -\frac{1}{2\sigma^2} [-2\mu_i^t x + \mu_i^t \mu_i] + \ln p(\omega_i)$$

Thus we obtain the equivalent linear discriminant functions

$$g_i(x) = \mathbf{w}_i^t x + w_{i0}$$

where

$$\mathbf{w}_i = \frac{1}{\sigma^2} \mu_i, w_{i0} = -\frac{1}{2\sigma^2} \mu_i^t \mu_i + \ln p(\omega_i)$$

if $g_i(x) > g_j(x)$, then $x \in \text{class } i$

if $g_j(x) > g_i(x)$, then $x \in \text{class } j$

Case 1: Decision Boundary Equation

$$g_i(x) = g_j(x) \text{ or } g_i(x) - g_j(x) = 0$$

$$g_i(x) = \mathbf{w}_i^t \mathbf{x} + w_{i0}$$

$$g_j(x) = \mathbf{w}_j^t \mathbf{x} + w_{j0}$$

$$g_i(x) - g_j(x) = 0$$

$$\Rightarrow (\mathbf{w}_i - \mathbf{w}_j)^t \mathbf{x} + w_{i0} - w_{j0} = 0$$

$$\Rightarrow \frac{1}{\sigma^2} (\mu_i - \mu_j)^t \mathbf{x} - \frac{1}{2\sigma^2} \mu_i^t \mu_i + \ln p(\omega_i) + \frac{1}{2\sigma^2} \mu_j^t \mu_j - \ln p(\omega_j) = 0$$

$$\Rightarrow (\mu_i - \mu_j)^t \mathbf{x} - \frac{1}{2} (\mu_i^t \mu_i - \mu_j^t \mu_j) + \sigma^2 \ln \frac{p(\omega_i)}{p(\omega_j)} = 0$$

$$\Rightarrow (\mu_i - \mu_j)^t \left[\mathbf{x} - \left\{ \frac{1}{2} (\mu_i + \mu_j) - \frac{\sigma^2}{\|\mu_i - \mu_j\|^2} \left[\ln \frac{p(\omega_i)}{p(\omega_j)} \right] (\mu_i - \mu_j) \right\} \right] = 0$$

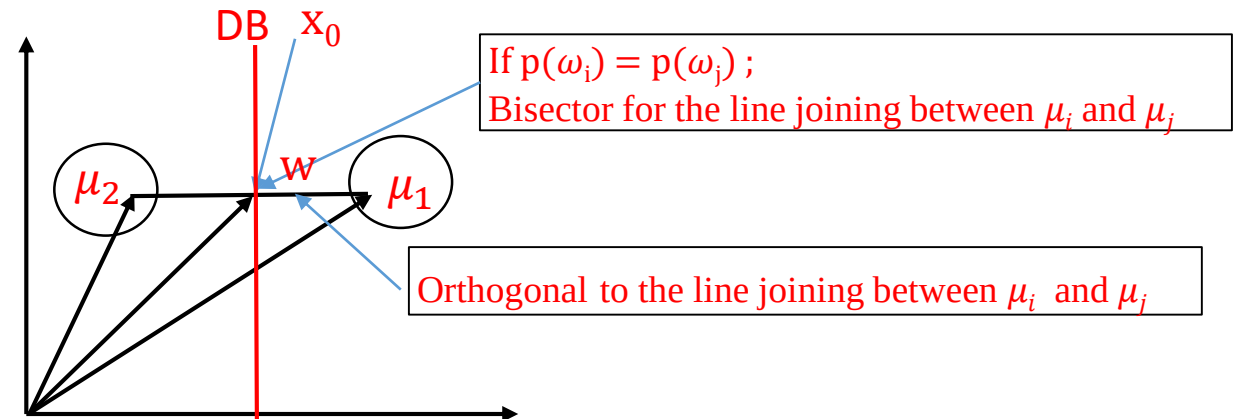
$$\Rightarrow \mathbf{w}^t (\mathbf{x} - \mathbf{x}_0) = 0$$

Case 1: Vector Interpretation

$$w = \frac{1}{2}(\mu_i - \mu_j) \qquad x_0 = \frac{1}{2}(\mu_i + \mu_j) - \frac{\sigma^2}{\|\mu_i - \mu_j\|^2} \left[\ln \frac{p(\omega_i)}{p(\omega_j)} \right] (\mu_i - \mu_j)$$

$$\text{If } p(\omega_i) = p(\omega_j); \qquad x_0 = \frac{1}{2}(\mu_i + \mu_j)$$

if σ^2 is small with respect to $\|\mu_i - \mu_j\|^2$, compact cluster



Case 1: Decision Boundary

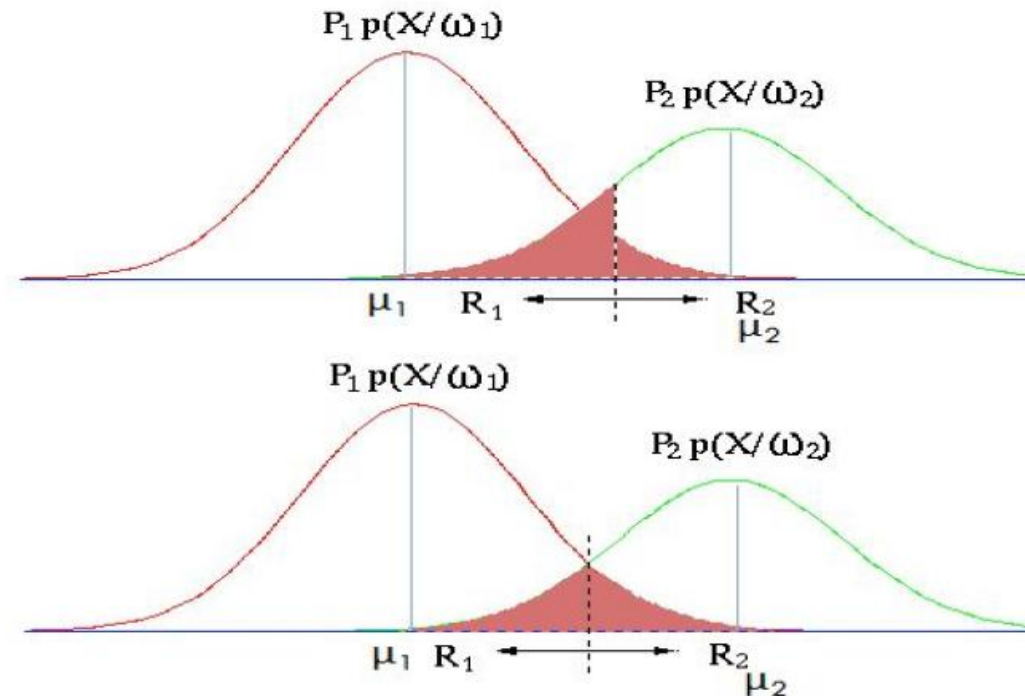


Figure: Decision Boundary

If $P(\omega_1) = P(\omega_2)$ It is on the point X_0

If $P(\omega_1) > P(\omega_2)$ The decision surface is away from μ_1

If $P(\omega_2) > P(\omega_1)$ The decision surface is away from μ_2

Example: Bivariate Normal density with $\sigma_1 = \sigma_2$

$$x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \quad \mu_i = \begin{bmatrix} \mu_{i1} \\ \mu_{i2} \end{bmatrix} \quad \Sigma = \begin{bmatrix} \sigma_1^2 & 0 \\ 0 & \sigma_2^2 \end{bmatrix} = \sigma^2 I$$

$$\Sigma^{-1} = \frac{1}{\sigma^2} I$$

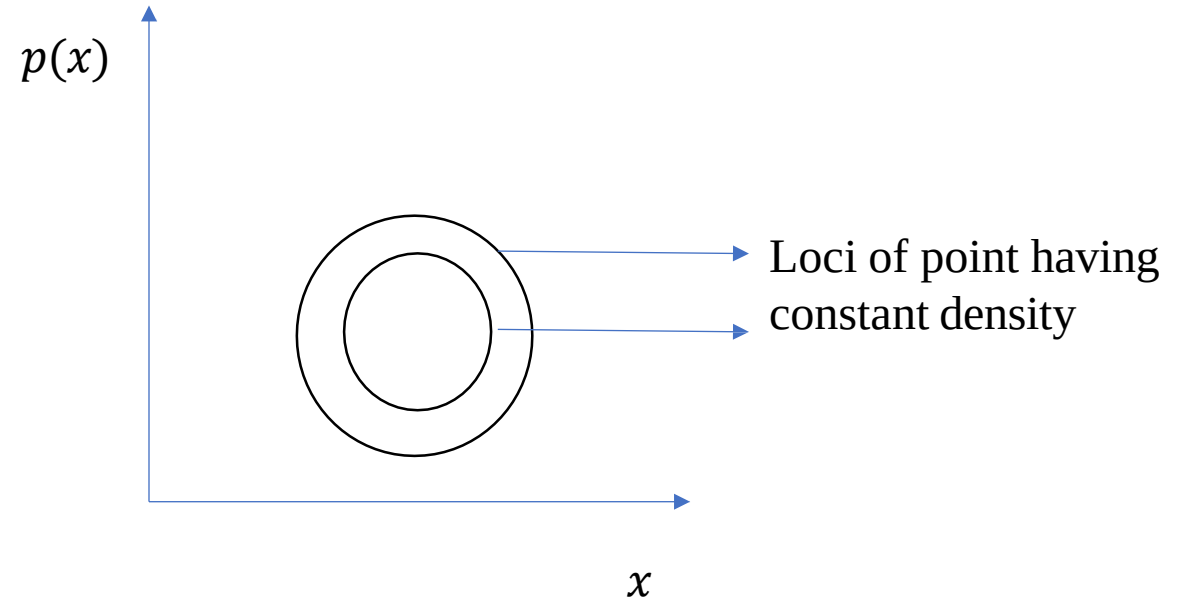
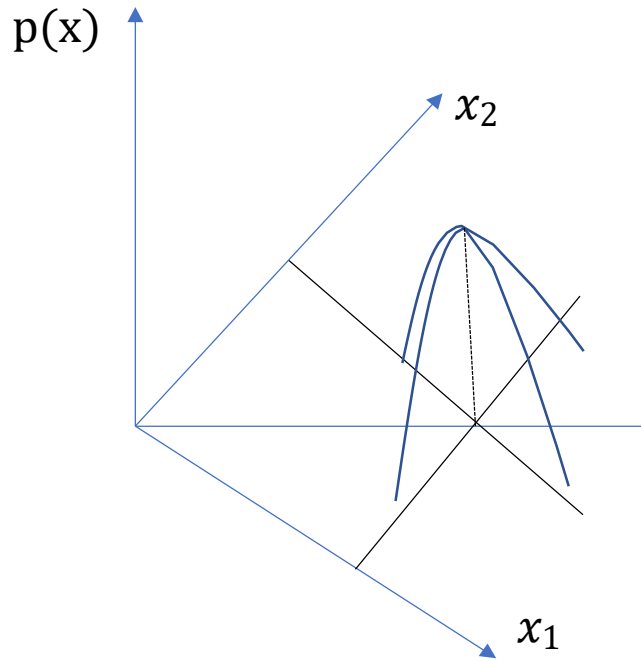
$$\ln p(x|\omega_i) = -\frac{1}{2} [x^t \Sigma_i^{-1} x - x^t \Sigma^{-1} \mu_i - \mu_i^t \Sigma^{-1} x + \mu_i^t \Sigma^{-1} \mu_i]$$

$$\ln p(x|\omega_i) = -\frac{1}{2} [x_1 \quad x_2] \frac{1}{\sigma^2} I \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} - 2[\mu_{i1} \quad \mu_{i2}] \frac{1}{\sigma^2} I \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + [\mu_{i1} \quad \mu_{i2}] \frac{1}{\sigma^2} I \begin{bmatrix} \mu_{i1} \\ \mu_{i2} \end{bmatrix}$$

$$\ln p(x|\omega_i) = -\frac{1}{2} \left\{ \frac{1}{\sigma^2} [x_1^2 + x_2^2] - \frac{2}{\sigma^2} [x_1 \mu_{i1} + x_2 \mu_{i2}] + \frac{1}{\sigma^2} [\mu_{i1}^2 + \mu_{i2}^2] \right\}$$

$$\ln p(x|\omega_i) = -\frac{1}{2\sigma^2} \{ (x_1^2 + x_2^2) - 2(x_1 \mu_{i1} + x_2 \mu_{i2}) + (\mu_{i1}^2 + \mu_{i2}^2) \}$$

Physical Interpretation



For bivariate function I want to trace the loci of constant density i.e. all value of x for which $p(x)$ is constant, those loci is nothing but circle.

Along each circle, all points are equally likely (same density). Points closer to the center circle are more likely to occur than points on outer circles.

Case 2: $\Sigma_i = \Sigma$

$$g_i(\mathbf{x}) = -\frac{1}{2} [(\mathbf{x} - \boldsymbol{\mu}_i)^t \Sigma_i^{-1} (\mathbf{x} - \boldsymbol{\mu}_i)] + \ln p(\omega_i)$$

$$= -\frac{1}{2} [(\mathbf{x}^t - \boldsymbol{\mu}_i^t) \Sigma_i^{-1} (\mathbf{x} - \boldsymbol{\mu}_i)] + \ln p(\omega_i)$$

$$= -\frac{1}{2} [(\mathbf{x}^t \Sigma_i^{-1} - \boldsymbol{\mu}_i^t \Sigma_i^{-1}) (\mathbf{x} - \boldsymbol{\mu}_i)] + \ln p(\omega_i)$$

$$= -\frac{1}{2} (\mathbf{x}^t \Sigma_i^{-1} \mathbf{x} - \boldsymbol{\mu}_i^t \Sigma_i^{-1} \mathbf{x} + \boldsymbol{\mu}_i^t \Sigma_i^{-1} \boldsymbol{\mu}_i - \mathbf{x}^t \Sigma_i^{-1} \boldsymbol{\mu}_i) + \ln p(\omega_i) \Rightarrow (1)$$

$\mathbf{x}^t \Sigma_i^{-1} \mathbf{x}$ is same for all classes and hence ignored

$$= -\frac{1}{2} (-2\boldsymbol{\mu}_i^t \Sigma_i^{-1} \mathbf{x} + \boldsymbol{\mu}_i^t \Sigma_i^{-1} \boldsymbol{\mu}_i) + \ln p(\omega_i)$$

$$\boldsymbol{\mu}_i^t \Sigma_i^{-1} \mathbf{x} == \mathbf{x}^t \Sigma_i^{-1} \boldsymbol{\mu}_i$$

$$= \boldsymbol{\mu}_i^t \Sigma_i^{-1} \mathbf{x} - \frac{1}{2} [\boldsymbol{\mu}_i^t \Sigma_i^{-1} \boldsymbol{\mu}_i] + \ln p(\omega_i)$$

$$\Sigma \text{ is arbitrary} \Rightarrow \Sigma_1 = \Sigma_2 = \begin{bmatrix} \sigma_1^2 & \sigma_{12} \\ \sigma_{21} & \sigma_2^2 \end{bmatrix}$$

x_1 and x_2 are not necessarily independent

Σ_i is the same for all different classes

The samples are clustered in hyper ellipsoidal of same shape and size

$\sigma_{12} = \sigma_{21}$, hence symmetry

$$g_i(\mathbf{x}) = \mathbf{w}_i^T \mathbf{x} + w_{i0} \Rightarrow \text{linear equation}$$

Where

$$w_i = \Sigma_i^{-1} \boldsymbol{\mu}_i$$

$$w_{i0} = -\frac{1}{2} [\boldsymbol{\mu}_i^t \Sigma_i^{-1} \boldsymbol{\mu}_i] + \ln p(\omega_i)$$

Case 2: Decision Boundary

What will be the nature of the decision boundary that separates the two classes ω_i and ω_j ?

$$g_i(\mathbf{x}) - g_j(\mathbf{x}) = 0$$

By deriving as like previous, it turned to

$$\mathbf{w}^t (\mathbf{x} - \mathbf{x}_0) = 0$$

Where,

$$\mathbf{w} = \Sigma^{-1} (\mu_i - \mu_j)$$

$$\mathbf{x}_0 = \frac{1}{2} (\mu_i + \mu_j) - \frac{1}{(\mu_i - \mu_j)^t \Sigma^{-1} (\mu_i - \mu_j)} \ln \frac{p(\omega_i)}{p(\omega_j)} (\mu_i - \mu_j)$$

squared Mahalanobis distance

Special Case: $\sigma_1^2 \neq \sigma_2^2$ of $\Sigma_i = \Sigma$

$$\Sigma = \begin{bmatrix} \sigma_1^2 & \sigma_{12} \\ \sigma_{21} & \sigma_2^2 \end{bmatrix}$$

If all samples are statistically independent,

$$\sigma_{12} = \sigma_{21} = 0$$

then

$$\Sigma = \begin{bmatrix} \sigma_1^2 & 0 \\ 0 & \sigma_2^2 \end{bmatrix}$$

Case 3: $\Sigma_i \neq \Sigma$

Data are not statistically independent.

It is more general case

Σ_i is arbitrary; different classes have different covariance matrix; $\Sigma_i \neq \Sigma_j$

From (1), We cant ignore anything here because of Σ_i is arbitrary in nature

$$g_i(\mathbf{x}) = -\frac{1}{2}[\mathbf{x}^t \Sigma_i^{-1} \mathbf{x} - \boldsymbol{\mu}_i^t \Sigma_i^{-1} \mathbf{x} - \mathbf{x}^t \Sigma_i^{-1} \boldsymbol{\mu}_i + \boldsymbol{\mu}_i^t \Sigma_i^{-1} \boldsymbol{\mu}_i] + \ln p(\omega_i) - \frac{1}{2} \ln |\Sigma_i|$$

$$g_i(\mathbf{x}) = -\frac{1}{2}[\mathbf{x}^t \Sigma_i^{-1} \mathbf{x} - 2\boldsymbol{\mu}_i^t \Sigma_i^{-1} \mathbf{x} + \boldsymbol{\mu}_i^t \Sigma_i^{-1} \boldsymbol{\mu}_i] + \ln p(\omega_i) - \frac{1}{2} \ln |\Sigma_i|$$

$$g_i(\mathbf{x}) = \mathbf{x}^t A_i \mathbf{x} + B_i^t \mathbf{x} + C_{io}$$

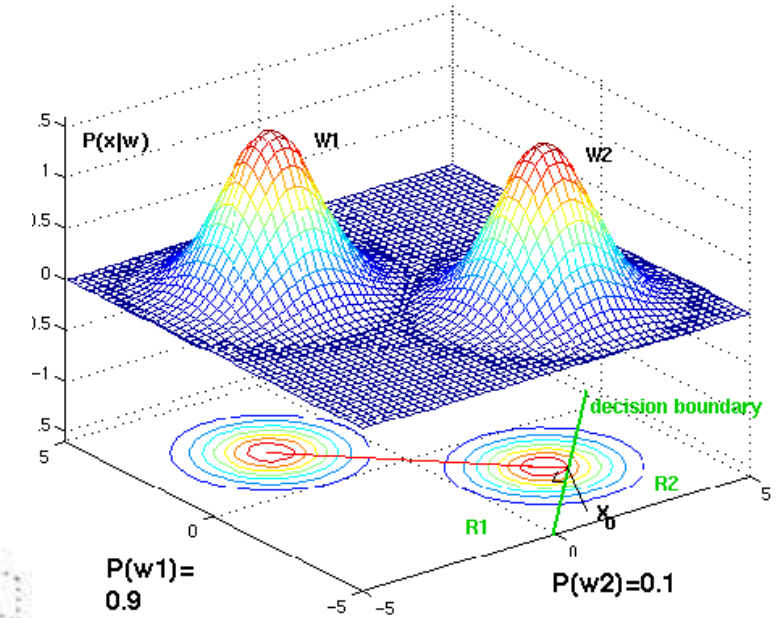
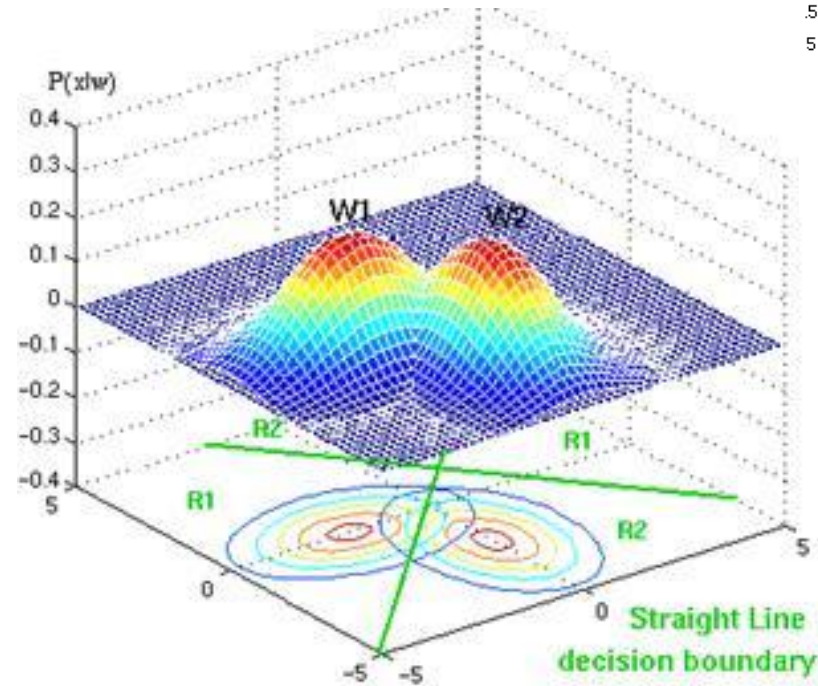
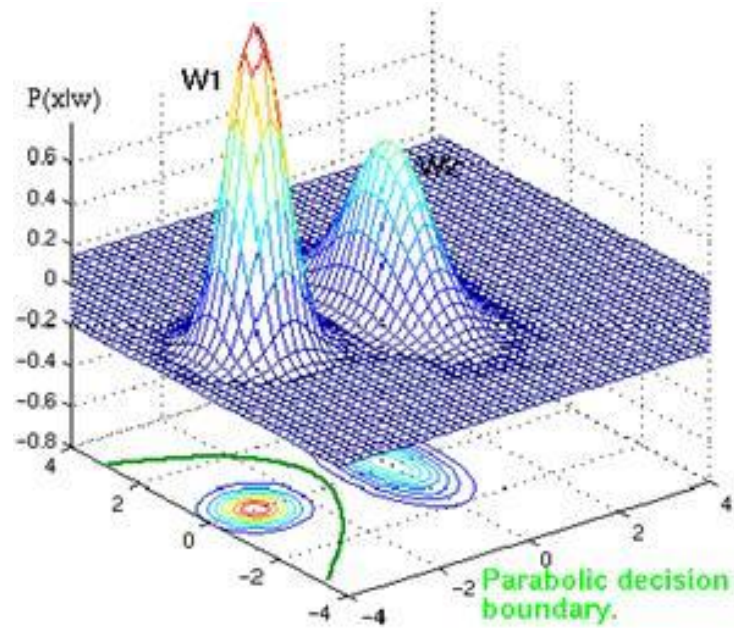
$$\text{where, } A_i = -\frac{1}{2} \Sigma_i^{-1}$$

$$B_i = \Sigma_i^{-1} \boldsymbol{\mu}_i$$

$$C_{io} = -\frac{1}{2} \boldsymbol{\mu}_i^t \Sigma_i^{-1} \boldsymbol{\mu}_i + \ln p(\omega_i) - \frac{1}{2} \ln |\Sigma_i|$$

The decision surface is quadratic hyperplane

Bivariate Normal density for two classes



Thank You